

Knowledge Mobilization Plan (max. 1 page)

Appropriate Knowledge Mobilization (KM) strategies will be used to reach key audiences: 1) members of the public; 2) transportation planners and practitioners; and 3) academic and scientific peers. The creation of transparent, research-based tools will improve planning processes and outcomes, and support fluid participation in planning processes by a variety of stakeholders.

Public and practitioners (Years 1-5): A key aspect of the proposed work is the use of a codesign framework. In codesign there is a continuous and multilateral flow of knowledge during every step of the project. In addition to the involvement of members of the public and practitioners in this endeavor, we will expand the diffusion of this research by writing general reports in layperson language to facilitate knowledge transfer. These reports will take the form of policy briefs, user's manuals, and sample case reports related to the release of the tools. These will be posted publicly via a website that will be hosted on GitHub Pages. Members of the research team have experience developing and releasing available materials for public use with this medium (59,60). Communication with the public and practitioners will begin immediately through the codesign framework.

Academic (Years 2-5): This research proposal includes 4 research students (2 Master's and 2 PhDs), two post-doctoral researchers, five professors, and one international collaborator. The goal is for each Master's student to lead at least one paper for a national conference and then publish in an international peer-reviewed journal. PhD students will aim to publish three peer-reviewed journal articles. The post-doctoral fellows will lead one research paper each. Thus, 10 papers would be led by trainees. This project primarily has implications for planning in urban regions. Accordingly, target journals for this research include *Transportation Research A: Policy and Practice*; *Public Transportation*; *Transport Policy*; *Journal of Transport Geography*; and *Transportation*. Pre-print versions of papers will be published in their GitHub repositories. Conferences will target audiences with both researchers and practitioners. Further knowledge diffusion will be achieved via workshops in relevant conferences. Communication with academic audiences will begin in year 2 of the project when initial results can be reported.

Open-Science Statement: The Tri-Council system in Canada (i.e., CIHR, NSERC, SSHRC) is currently phasing in an Open Access policy for the research it funds. Open Access of publications and data are only one facet of open science, since encouraging quality of work throughout the research process requires an effort to share the data-collection process, methodology, and analysis as well (66). Open science can encourage and increase engagement from members of the public regarding ongoing research and any relevant issues that can emerge (61). In this respect, members of the research team have already a proven record of creating open, reproducible science, not only by publishing preprints publicly and sharing data, but by sharing every aspect of research, from the stage of data preparation and up to reproducible computational notebooks. Examples of data sharing by members of the research team include the creation of several open-data products (62) such as ActiveCA package (63) and TTS2016R package (59). An example of equity-based research in transportation that illustrates the mode of sharing pre-prints along with reproducible notebooks is Desjardins et al. (64). Materials such as this, shared publicly on websites like GitHub, allow for thorough verification of the research, to ensure it does not contain major errors that could impact the validity of the results (65). Open-science practices like this also speed up the process of discovery, by making it easy for other researchers to access the base work and modify it to accomplish whatever is required for their own specific work. The members of the research team are experienced open-science researchers and are fully committed to following Tri-Agency policy on Open Access, and more broadly to continue producing high-quality research that is open, transparent, and reproducible.